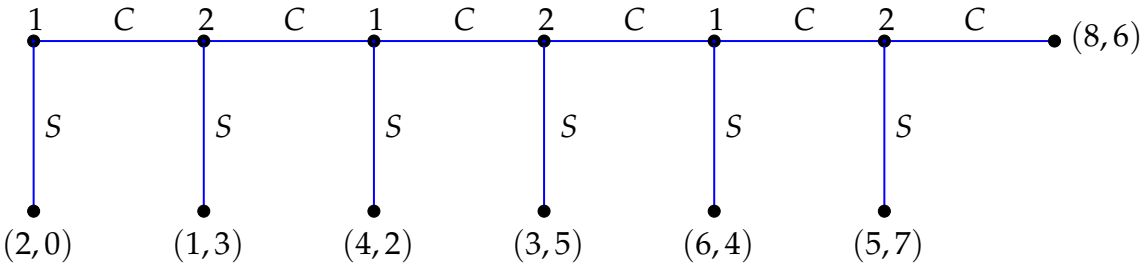
MTH636/SDS636 (2026-27, ODD SEMESTER)

PRACTICE PROBLEM SET 4

1. In the centipede game, the two players move alternatingly. On each move, a player can stop (S) or continue (C). On any move, a player is better off stopping the game than continuing if the other player stops immediately afterward, but is worse off stopping than continuing if the other player continues, regardless of the subsequent actions. The game ends after a finite number of periods. Consider an example of this game below.



   (a) Determine the backward induction or subgame perfect equilibrium of this game. What is the associated outcome?

   (b) Show that there are other Nash equilibria, but that these always result in the same outcome as the subgame perfect equilibrium.

2. Allen and Rick need to divide \$100 between them as follows: first Allen suggests an integer x between 0 and 100 (which is the amount of money he wants for himself). Rick, on hearing the suggested amount, decides whether to accept or reject. If Rick accepts, the payoff of the game is $(x, 100 - x)$: Allen receives x dollars, and Rick receives $100 - x$ dollars. If Rick chooses to reject, neither player receives any money.

   (i) Describe this situation as an extensive-form game.

(ii) What is the set of pure strategies each player has?

(iii) Show that any result $(a, 100 - a)$, $a \in \{0, 1, \ldots, 100\}$, is a Nash equilibrium payoff. What are the corresponding equilibrium strategies?

(iv) Find all the subgame perfect equilibria (in pure strategies) of this game.

3. Bill asks Al to choose heads or tails. After Al has made his choice (without disclosing it to Bill), Bill flips a coin. If the coin falls on Al's choice, Al wins. Otherwise, Bill wins. Depict this situation as a game in extensive form.

4. Consider the following game. Player I has the opening move, in which he chooses an action from the set $\{L, R\}$. A lottery is then conducted, with either λ or ρ selected, both with probability $\frac{1}{2}$. Finally, Player II chooses either l or r. The outcomes of the game are not specified. Depict the game tree associated with the extensive-form game in each of the following situations:

   (a) Player II, at his turn, knows Player I's choice, but does not know the outcome of the lottery.

   (b) Player II, at his turn, knows the outcome of the lottery, but does not know Player I's choice.

   (c) Player II, at his turn, knows the outcome of the lottery only if Player I has selected L.

   (d) Player II, at his turn, knows Player I's choice if the outcome of the lottery is λ, but does not know Player I's choice if the outcome of the lottery is ρ.

   (e) Player II, at his turn, does not know Player I's choice, and also does not know the outcome of the lottery.

5. In the following game (Figure 1), the root is a chance move, Player I has three information sets, and the outcome is the amount of money that Player I receives.

   (a) What does Player I know in each of his information sets, and what does he not know?

   (b) What would you recommend Player I to play, assuming that he wants to maximize his expected payoff? Justify your answer.

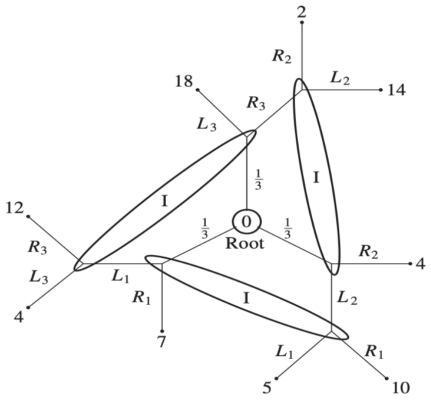


Figure 1

6. Let h and h' be two distinct information sets (they may both be the information sets of the same player, or of two different players). h will be said to precede h' if there exist a vertex $x \in h$ and a vertex $x' \in h'$ such that the path from the root to x' passes through x

   (a) Find a two-player game with chance moves, where both players have perfect recall and there exist two information sets h and h' such that h precedes h', and h' precedes h.

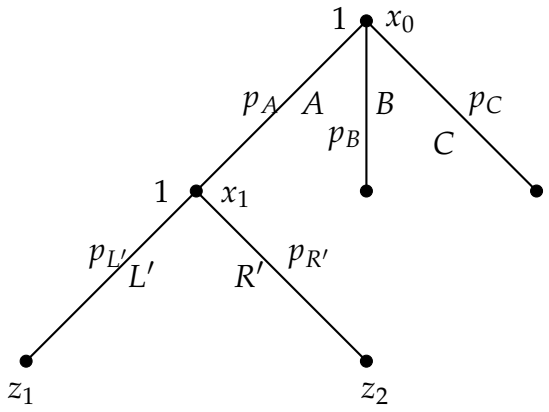
   (b) Find a three-player game without chance moves, where all players have perfect recall and there exist two information sets h and h' such that h precedes h', and h' precedes h.

7. Suppose $\sigma^\star$ is an NE of a finite extensive-form game in which every chance action has strictly positive probability, and suppose each $\sigma_i^\star$ assigns positive probability to every pure strategy of Player i. Prove or disprove: $\sigma^\star$ is subgame perfect.

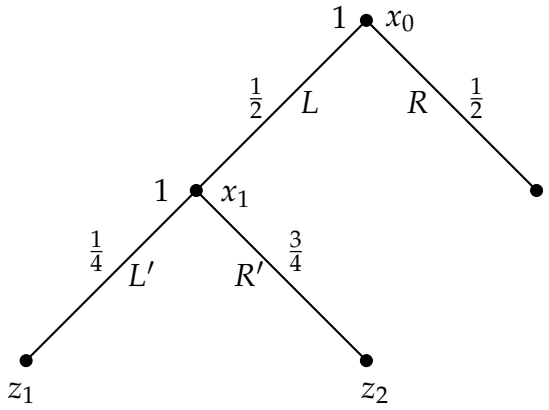
8. Determine all the behavior strategies that are outcome equivalent with the following mixed strategy

AL'	AR'	BL'	BR'	CL'	CR'
0.5	0	0.2	0.1	0.1	0.1

   in the following one-player extensive form structure:



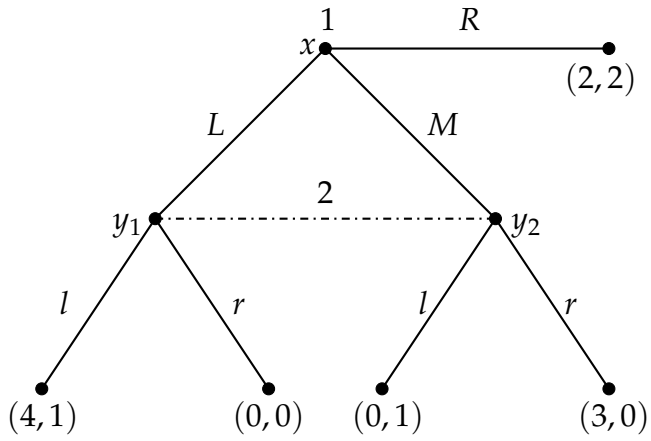
9. Determine all mixed strategies that are outcome equivalent with the behavior strategy represented in the following one-player extensive form structure:



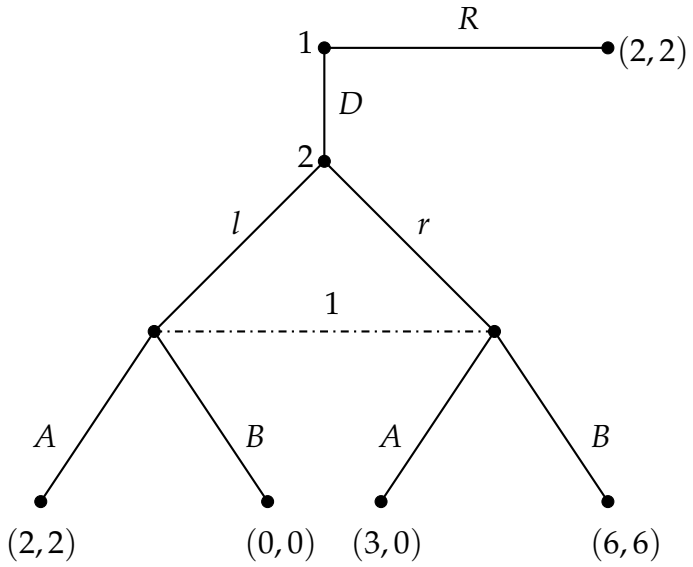
10. Consider the following extensive form game below.

    (a) Determine the strategic form of this game,

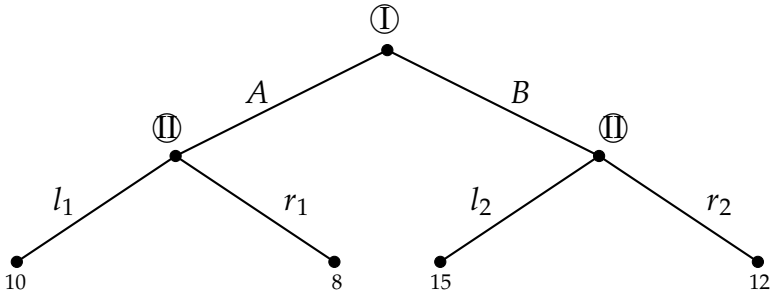
    (b) Determine all Nash and subgame perfect Nash equilibria of this game.



11. Compute all Nash and Subgame Prfect equilibria in the following game.



12. Consider the following two-player zero-sum game in extensive form game:



(a) Find an optimal strategy for each player by applying backward induction.

(b) Describe this game in strategic form.

(c) Find all the optimal strategies of the two players.

(d) Explain why Player II has optimal strategies other than the strategy obtained by backward induction.

13. Consider a one-player extensive-form game. Player 1 first chooses either L or R. If R is chosen, the game ends. If L is chosen, Player 1 moves again and chooses either L' or R'. All decision nodes are singleton information sets.

Consider the following two mixed strategies:

$$\sigma = \frac{1}{2}(LL') + \frac{1}{2}(RR')$$

and

$$\sigma' = \frac{1}{2}(LR') + \frac{1}{2}(RL').$$

Notice that under either mixed strategy, the marginal probability of planning to choose L at the first information set is $1/2$, and the marginal probability of planning to choose L' at the second information set is also $1/2$.

(a) Are σ and σ' outcome equivalent? Justify your answer.

(b) Determine the behavior strategy that is outcome equivalent to σ.

(c) Determine the behavior strategy that is outcome equivalent to σ'.

(d) Explain why the marginal probabilities of the actions under a mixed strategy need not coincide with the corresponding probabilities in an outcome-equivalent behavior strategy.